

Representation Bias, Correction Transfer, and Resolution Sensitivity in 3D Mitochondrial Morphometry

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Abstract—Mitochondrial shape measurements can change with representation and processing. We evaluate representation bias and correction transfer in 2,720 development and 2,728 previously unused objects from the 3D Mitochondria Shape Library for Optical Microscopy (3DMSL). A separate resolution study uses 550 rat-cortex objects from MitoEM-R. Development occupancy volumes exceeded mesh volumes by a mean of 3.665% despite an intraclass correlation coefficient of 0.994. Among 57 accepted boundary fits, the median outward label-transition displacement was 0.00304 normalized units. An exploratory pipeline reimplement, with five pilot objects retrospectively excluded, showed a mean volume-error increase of 1.571 percentage points in 55 objects when the depth offset increased from 0 to 1.5 voxels. A frozen occupancy-only regression reduced second-batch median absolute percentage error from 3.481% to 0.664% and root mean square percentage error from 5.743% to 1.743%. Coverage of its 2.661% calibrated bound fell from 96.2% internally to 92.1%; coverage was 95.1% outside a pre-existing low-occupancy flag and 49.2% within it. In MitoEM-R, coarsening in-plane spacing from 8 to 24 nm changed median volume by -0.08% , surface area by -10.60% , and sphericity by $+11.76\%$, while sphericity ranks remained highly correlated ($\rho = 0.994$). Grid-phase checks qualified volume preservation. Numerical precision, absolute agreement, rank stability, and uncertainty transfer require separate assessment. These results establish within-resource processing effects, not biological invariance or cross-pipeline correction portability.

Index Terms—Mitochondrial morphometry, representation bias, measurement reliability, electron microscopy, occupancy networks, conformal calibration, resolution sensitivity.

I. INTRODUCTION

Mitochondrial morphology is a regulated phenotype shaped by fission, fusion, and interactions with the cellular environment [1]. Its relationship with bioenergetics is bidirectional: structural perturbations can accompany changes in respiration, and metabolic interventions can reshape the mitochondrial network [2]. A morphological difference therefore has to be interpreted at two levels. An apparent change in volume, surface area, or sphericity can reflect altered organelle structure, but it can also arise from segmentation, meshing, sampling, or resolution.

Work that combines structural and functional measurements makes the biological question more tractable. Faitg et al. developed a method that pairs electron microscopy (EM) morphology with cytochrome c oxidase histochemistry in the same tissue [3]. Tseng et al. combined live-cell imaging, network

analysis, and biophysical modeling in pancreatic beta cells [4]. More recent studies paired single-organelle morphology with fluorescent functional probes [5], or shape phenotyping with metabolic measurements for colorectal cancer cell classification [6]. Functional readouts carry their own assay-dependent uncertainty [7]. The structural side of such studies needs the same scrutiny.

This study treats representation conversion as a measurement problem. Its primary question is whether quantities derived from the same annotated object remain comparable after conversion, correction, or regridding. Biological interpretation requires additional evidence beyond that computational agreement.

We ask four questions. First, can a high whole-sample agreement coefficient conceal systematic volume bias? Second, is the measured boundary discrepancy consistent with the proposed preprocessing mechanism? Third, can a correction using occupancy-derived features transfer to previously unused shapes without access to a mesh at prediction time, and does its uncertainty bound transfer with it? Fourth, how does voxel resolution affect the absolute values and rankings of descriptors that may later enter a structure-function model?

The contributions are: (1) a coordinate-consistent audit that separates sampling precision from representation agreement; (2) boundary measurements and an exploratory controlled reimplement supporting a depth-offset contribution to volume inflation; (3) frozen-model evaluation across two disjoint-ID batches within 3DMSL, with markedly elevated failure risk in a pre-existing low-occupancy subgroup; and (4) an all-eligible-object MitoEM-R test showing that stable ranks can coexist with changes in absolute descriptors and topology. Additional exploratory sensitivity analyses quantify the effects of grid phase, tie handling, and normalized geometric scale. The MitoEM-R analysis tests descriptor sensitivity independently of the 3DMSL correction; it is not a cross-resource validation of that correction.

II. RELATED WORK

A. Structural measurements and functional interpretation

The studies summarized in Table I combine structural and functional information at different scales. Tissue histochemistry, live-cell network dynamics, single-organelle probes, and cell-level metabolic profiling address different endpoints. These approaches motivate paired measurements, but do not establish a universal mapping from shape to function. Their

biological endpoints are not directly comparable with the representation-error percentages reported here.

B. Automated phenotyping tools

MiNA measures mitochondrial network skeletons [8]. Mito-Graph reconstructs three-dimensional (3D) surfaces and estimates volume [9]. Nellie extends automated organelle analysis to segmentation, tracking, and hierarchical features in 2D and 3D live-cell microscopy [10].

Classification tools address complementary questions. MitoClass assigns network morphologies to fragmented, intermediate, or elongated categories with a convolutional neural network [11]. MitoTex provides texture-based characterization [12]. These tools produce descriptors from a chosen representation; the agreement of such descriptors across representations of the same object is the question studied here.

C. Shape representations and public resources

3DMSL provides meshes, point clouds, implicit occupancy data, and simulated optical microscopy images for more than 27,000 mitochondrial shapes [13]. The upstream FIB-SEM resource is archived as EMPIAR-10791 [14]. Its originating study investigated liver subcellular architecture and metabolic homeostasis [15]. The 3DMSL repository identifies obese (ob/ob) mouse liver as the source used for the shape library [16]. Its occupancy labels follow the Occupancy Networks preprocessing [17], in which query points are labeled against a watertight surface produced by depth-map fusion [18]. MitoEM provides large-scale instance-labeled EM volumes [19], and the subsequent challenge analysis documents the importance of annotation quality and evaluation design [20]. Derived representations of one annotation are not independent measurements of biological truth, but they allow controlled processing studies in which object identity is fixed.

D. The measurement gap

Table I places this study relative to representative approaches. It compares scope and evidence, not benchmark performance; no cited model was rerun on a common benchmark. The specific question addressed here is whether a quantitative descriptor remains comparable after representation conversion and whether a correction and its uncertainty bound transfer together. The contribution is the empirical characterization of these distinct failure modes in paired mitochondrial representations, rather than a new agreement statistic, regression algorithm, or conformal-prediction theorem.

III. MEASUREMENT FRAMEWORK

Let S_i denote the underlying structure of object i and let T_r denote processing into representation r . A descriptor is observed as $m_{ir} = g_r(T_r(S_i))$, so m_{ir} and m_{is} may differ even when the object is unchanged. Our experiments hold object identity fixed and vary representation or processing. The raw 3DMSL mesh serves as the computational reference; it is not treated as an error-free biological measurement.

For mesh volume V_m and occupancy volume V_o , the signed percentage error is

$$e = 100 \left(\frac{V_o}{V_m} - 1 \right). \quad (1)$$

We report mean signed error, median absolute percentage error (MdAPE), and root mean square percentage error (RMSE). Spearman rank correlation assesses ordering. Agreement is summarized with the two-way, absolute-agreement, single-measure intraclass correlation coefficient, ICC(A,1) [21], interpreted following [22]. ICC depends on between-object variability relative to disagreement; a wide size range can yield a high ICC despite practically important object-level errors. We therefore also report Bland–Altman bias and limits of agreement [23] and the percentage error, $100 \times 1.96 \text{SD}_\Delta / \bar{M}$, where $\bar{M} = n^{-1} \sum_i (V_{mi} + V_{oi})/2$ [24].

For an approximately uniform outward boundary displacement t that is small relative to local geometric dimensions, the first-order volume change and the implied normalized displacement are

$$\Delta V \approx A_m t, \quad t_{\text{norm}} \approx \frac{V_o - V_m}{A_m s}, \quad (2)$$

where A_m is mesh area and s is the stored normalization scale. For constant normalized displacement, the corresponding prediction is

$$e \approx 100 t_{\text{norm}} \frac{A_m s}{V_m} = 100 t_{\text{norm}} \frac{A_m}{V_m^{2/3}} \frac{s}{V_m^{1/3}}. \quad (3)$$

Thus, dimensionless shape complexity $A_m/V_m^{2/3}$ alone omits a potentially variable scale-to-size factor. We report its association with error descriptively and additionally test $A_m s/V_m$ as a mechanistic predictor.

The repeatability coefficients, residual bounds, and candidate error margins estimated below describe the structural measurement process only. They have no established equivalence to functional-transition thresholds.

IV. DATASETS AND EXPERIMENTAL DESIGN

A. Dataset provenance and units of analysis

We analyzed two public resources, 3DMSL and MitoEM-R (Table II). The two 3DMSL ZIP files are batches within one resource, not independent biological cohorts. Their IDs are disjoint. A post-audit check found no identical raw-mesh file hashes across batches; this does not establish source-animal independence or exclude geometrically equivalent objects encoded differently. EMPIAR-10791 is the upstream source of 3DMSL, not a third analyzed dataset.

E4 and E10 use both 3DMSL batches, while E8 and E9 reuse subsets of the E5 objects; rows of Table II should not be summed. The 5,448 disjoint 3DMSL IDs are not treated as independent animals. An initial 40-object MitoEM analysis and an initial 400-object imaging analysis were superseded by the 550-object and 300-object analyses and are not counted as additional evidence.

TABLE I
REPRESENTATIVE APPROACHES AND THE MEASUREMENT QUESTION ADDRESSED BY THIS STUDY.

Study or resource	Main approach	Evidence or endpoint	Relationship to this study
Faitg et al. [3], 2020	Combined EM morphology and cytochrome c oxidase labeling method	Integrated structural and functional assessment	Motivates paired biological measurements; our endpoint is representation error
MitoEM [19], 2020; challenge analysis [20], 2023	Large-scale 3D EM instance segmentation	Annotated objects and segmentation evaluation	Supplies real-EM labels; we measure downstream descriptor sensitivity
Tseng et al. [4], 2024	Live-cell imaging with network and biophysical modeling	Metabolic coupling of mitochondrial dynamics	Addresses biological coupling; we characterize measurement prerequisites
Nellie [10], 2025	Automated 2D/3D organelle segmentation and feature extraction	Hierarchical morphology and motility features	Produces descriptors whose representation sensitivity we quantify
MitoClass [11], 2025	CNN-based morphology classification	Fragmented, intermediate, and elongated networks	Categorical phenotyping rather than paired volume-error calibration
Ding et al. [5], 2025	Fluorescent probes with AI-based organelle analysis	Joint morphology and functional biomarkers	Direct multimodal evidence absent from our structural datasets
MitoTex [12], 2026	Texture extraction and classification interface	Texture-based phenotyping	Complements geometric descriptors
Charrasse et al. [6], 2026	Morphological and metabolic profiling	Colorectal cancer cell classification	Illustrates added information from combined measurements
3DMSL [13], 2026 article	Paired 3D shapes and synthetic microscopy	Resource for reconstruction and imaging methods	Enables controlled comparison of representations of the same object
Present study	Bias audit, boundary mechanism, frozen correction, subgroup and resolution tests	Object-level errors and transferred error-bound coverage	Quantifies representation sensitivity; does not infer mitochondrial function

TABLE II
DATASETS, ANALYZED SUBSETS, AND EXPERIMENT PARTICIPATION. E1–E10 ARE THE ANALYTICAL EXPERIMENTS IN TABLE III; S1 IS THE SUPPLEMENTARY IMAGING ANALYSIS IN THE APPENDIX.

Resource and citation	Analyzed subset	Objects or images used	Experiments
3DMSL dataset [25]	Development archive 10_24553_27272.zip; IDs 24553–27272	2,720 objects; internal split 1,630/540/550; repeatability 400; boundary test 60; offset test 55; hybrid check 10; imaging 300	E1–E5, E8–E10, S1
3DMSL dataset [25]	Frozen validation archive 1_1_2729.zip; IDs 1–2728	2,728 objects; 0 exclusions; 0 development-ID overlaps	E4, E10
MitoEM-R [19]	Rat cortex validation slices 400–499; $30 \times 8 \times 8$ nm voxels	1,203 instance IDs present in the slab; 550 eligible complete objects, all meshed	E6, E7
MitoEM-R [19]	Training slices 0–29	697 instance IDs present; 95 eligible; 4 meshed	Technical pilot only

B. Experiment-to-claim mapping

The experiment labels in Table III organize completed analyses; they do not imply prospective registration. E8–E10 are exploratory follow-ups, not additional independent cohorts. Development exploration preceded the internal train/calibration/test split. The second-batch evaluation used saved model parameters without refitting. Boundary and distribution-shift diagnostics are exploratory analyses that explain or qualify the primary results.

C. Structural processing and quality control

The 3DMSL audit excludes archive metadata files and verifies the mesh, point-cloud, and occupancy files for each object. Point-cloud and occupancy coordinates are transformed with $\mathbf{x}_{\text{raw}} = s \mathbf{x}_{\text{norm}} + \text{loc}$, and bit-packed occupancy labels are unpacked before processing. Each archive stores 100,000 uniformly distributed query points per object in a normalized box of ± 0.55 (padding 0.1), with labels computed against the watertight fused mesh [16]. The data article reports 10,000

points [13]; the released files and build script contain 100,000. The implemented occupancy volume estimate is the fraction of inside-labeled points multiplied by the product of the observed query-coordinate spans in raw units. This empirical-box convention is retained for consistency with the saved correction; it is distinct from using the nominal box width $1.1s$. Mesh volume is the absolute signed volume; mesh area and axis-aligned extents are recorded separately.

Load failures are hard exclusions. Four conditions are advisory flags: nonwatertight meshes; occupancy fractions below 0.02 (the low-occupancy flag); incomplete rendered views; and a raw, untrimmed point-cloud extent that differs from the mesh extent by more than 10% along any single axis. These flags were defined in the initial audit, before any correction model was fitted. Descriptor comparisons of point-cloud extent use the 1st-to-99th-percentile trimmed extent, which suppresses isolated outliers but also removes genuine extremal points. Both 3DMSL batches contained only watertight meshes.

TABLE III
EXPERIMENT-TO-CLAIM MAP.

ID	Experiment	Sample and comparison	Supported claim
E1	Integrity and representation agreement	2,720 development objects; mesh, occupancy, point cloud	Systematic disagreement despite high whole-sample agreement
E2	Occupancy sampling repeatability	400 development objects; 10 disjoint query-point folds; convergence analysis	Numerical precision is separate from representation offset
E3	Internal correction and calibration	1,630 training, 540 calibration, 550 test objects	Internal performance after exploratory development
E4	Frozen transfer and shift diagnostics	2,728 new objects; original model and bounds	Within-dataset transfer and location of coverage failures
E5	Label boundary characterization	60 development objects; distance-label logistic fits	Outward label-boundary offset consistent with the label pipeline
E6	Native MitoEM representation comparison	550 eligible objects from one rat cortex slab	Mask-to-mesh and point-sampling sensitivity on real EM
E7	In-plane resolution sensitivity	Same 550 objects; native, 16 nm, and 24 nm in-plane grids	Absolute descriptor shifts and rank stability under regridding
E8	Exploratory depth-offset intervention	55 non-pilot E5 objects; offsets 0, 0.75, 1.5; all 60 as sensitivity	Depth-offset contribution within the recreated pipeline
E9	Exploratory hybrid-label check	First 10 non-pilot E5 objects; offsets 0 and 1.5; field-band replacement	Limited sensitivity to label replacement in this subset
E10	Exploratory correction-model comparison	Same development split and 2,728-object second batch; fraction-only OLS and ridge	Model sensitivity after outcomes were inspected

D. Repeatability estimation

A simple random sample of 400 numerically ordered development IDs is selected without replacement with NumPy seed 0. The same generator subsequently permutes each object's stored queries into ten disjoint folds. The standard deviation of the ten fold-level volume estimates, divided by $\sqrt{10}$, estimates the standard deviation of the full-sample estimate under the independent-query model. The pairwise repeatability coefficient is $1.96\sqrt{2} \approx 2.77$ times this SD, expressed relative to the estimate. It describes differences between two independent numerical estimates, rather than a single estimate's error relative to a reference. A single-estimate normal 95% half-width would instead be 1.96 times the SD. These quantities exclude repeat scanning, segmentation, and biological variability. Random partitioning and marginal uniformity checks do not establish independence of query generation. A convergence analysis evaluates subsets from 1,000 to 100,000 queries.

E. Correction and calibration

The internal split is stratified by mesh-volume decile using seed 2026 and numerically ordered instance IDs; saved split identifiers are retained. Training data alone determine the regression coefficients and feature standardization, calibration data determine residual bounds, and test data provide internal performance summaries. Because earlier whole-batch exploration may have informed feature selection, the internal test is validation after exploration rather than prospective confirmation; the second batch was evaluated using the saved correction without refitting; subsequent analyses of that batch were exploratory.

Let B_q be the product of coordinate spans over all query points in raw units, p the fraction labeled inside, and B_{in} the bounding-box volume of only the inside-labeled points. The volume estimate is $V_o = pB_q$. The five predictors are V_o/B_{in} , p , $\log V_o$, $\log B_{\text{in}}$, and the ratio of maximum to minimum

coordinate span among the inside-labeled points. Thus, the fill predictor V_o/B_{in} and the query occupancy fraction p have different denominators and are not duplicate features. They describe bounding-box fill, sampling occupancy, scale, and elongation without requiring the reference mesh. Ordinary least squares supplies a transparent baseline; the study does not establish that this feature set or model class is optimal. It predicts the signed percentage error, predictions are clipped to $[-50, 100]\%$, and the corrected volume is

$$V_{\text{corrected}} = \frac{V_o}{1 + \hat{e}/100}. \quad (4)$$

The global correction substitutes the mean training-set percentage error for $\hat{e}$, and the uncorrected baseline retains V_o . Mesh values are used as development targets and for evaluation but are not predictors at deployment. The uncorrected and global-offset baselines distinguish the benefit of any adjustment from that of an object-specific adjustment. The separate mesh-informed geometric regression is a mechanism diagnostic, not a deployment competitor with the same available inputs.

For each method, calibration uses the absolute corrected-volume percentage error

$$a_i = \left| 100 \left(\frac{\hat{V}_i}{V_{mi}} - 1 \right) \right|. \quad (5)$$

For the occupancy-only correction, the signed residual is $(e_i - \hat{e}_i)/(1 + \hat{e}_i/100)$, rather than $e_i - \hat{e}_i$. The saved bound q is the $[(n_{\text{cal}} + 1)0.95]$ th ordered score: the 514th of 540 calibration scores here. For a general smaller calibration set with this index greater than n_{cal} , the conservative bound is infinite. For $q < 100\%$, inversion gives a reference-volume prediction interval

$$\left[\frac{\hat{V}}{1 + q/100}, \frac{\hat{V}}{1 - q/100} \right]. \quad (6)$$

The split-conformal construction [26] provides marginal coverage under the required exchangeability and fitting conditions, not conditional coverage for each subgroup [27]. Batch shift can violate exchangeability [28]. Earlier full-batch feature exploration also prevents treating the internal analysis as a clean prospective coverage guarantee. We report empirical coverage at the saved bound.

Frozen evaluation preserves the development feature means and scales, coefficients, clipping rule, global offset, and calibrated bounds. In the recorded evaluation workflow, model artifacts are copied and hashed before evaluation and the hashes are rechecked afterward. This checks file stability during that run; it does not independently establish the chronology of every analytical decision. No second-batch outcome was used to update the retained correction in the original frozen evaluation. Later model comparisons are separately labeled exploratory. Separate descriptive regressions on the second batch characterize error associations and are not refits of the deployed model.

F. Boundary and subgroup diagnostics

The boundary analysis uses 60 development objects sampled without replacement with seed 790; their recorded IDs are reused in the exact-distance refit. That refit uses query-sampling seed j for zero-based object index j . It analyzes the occupancy queries near the raw mesh in normalized coordinates. Points with an approximate distance below 0.02 (computed from the 32 nearest triangle centroids) are candidates, and up to 3,000 per object are retained. Their distances are then computed exactly, as the point-to-triangle distance over every triangle that could lie closer than the approximate value, with the inside/outside sign given by the generalized winding number [29]. The labels are fitted with the logistic model

$$P(\text{inside} \mid d) = \left[1 + \exp\left(\frac{d - t_{50}}{w}\right) \right]^{-1}, \quad (7)$$

where the midpoint t_{50} estimates the label-transition displacement and w its width. The model is fitted by maximum likelihood from five starting points. A fit is accepted only if the optimizer reports convergence, the scaled projected gradient is below 10^{-4} , neither parameter lies on a bound, the starts that reach the optimum agree on t_{50} to within 10^{-5} , each label class contains at least 20 points, and the classes overlap.

The measured displacement is compared with a heuristic reference from the documented label pipeline [16]. Meshes are scaled to largest extent 0.9, and depth maps from 100 views are fused at 256^3 resolution with a 1.5-voxel depth shift toward the camera in the public mesh-fusion implementation [30]. The corresponding full shift is $1.5/(256 \times 0.9) = 0.00651$ normalized units. Half of this, 0.00326, is an orientation-averaging heuristic, not a derived prediction for the complete fusion algorithm: visibility, weighting, depth-map morphology, discretization, and surface geometry can alter the final displacement. The released dataset's generating code version has not been confirmed. The controlled reimplementation below tests a candidate mechanism without claiming reproduction of the historical run.

Second-batch diagnostics compare shape distributions, implied offsets, low-occupancy prevalence, and coverage by shape-complexity decile. A development regression of error on shape complexity predicts the change in mean raw error expected from the change in mean shape complexity; this is an association-based decomposition. The worst-error analysis is post hoc. A post-audit diagnostic fits a zero-intercept model $e = \beta A_m s / V_m$ on development objects and applies the fitted slope unchanged to the second batch. It uses mesh information and serves only as a mechanistic diagnostic, not a deployment correction.

G. Exploratory controlled pipeline reimplementation

In E8, a subsequent NumPy implementation recreated rendering, depth-map erosion, truncated signed-distance fusion, and surface extraction, retaining 100 views and a 256^3 grid while varying only the depth offset (0, 0.75, and 1.5 voxels). Raw meshes, views, query points, and other settings were held fixed within object. Two coordinate conventions were evaluated: the literal half-voxel surface shift and a centred convention. Occupancy labels were obtained by trilinear interpolation of the fused signed-distance field, which is not generally identical to containment in the extracted mesh. E9 used the first ten non-pilot IDs at offsets 0 and 1.5 under the centred convention; the available records do not establish prospective selection. Generalized-winding-number containment replaced interpolated labels only where the absolute interpolated fused-field value was below $2/256$; labels elsewhere were retained. This hybrid check uses a field-value threshold, not a verified two-voxel Euclidean-distance band, and does not test mesh containment for every query point. Its near-surface agreement summaries use the separate sampled set defined relative to the raw mesh. Rendering was also reimplemented; agreement with released labels therefore measures reconstruction fidelity, not exact pipeline identity or biological accuracy.

The run reused the 60 E5 objects. We retrospectively excluded five engineering-pilot IDs (24558, 24611, 24853, 24898, 24925) inadvertently retained in the initial summary, leaving 55 objects; all 60 were retained for sensitivity analysis. Exclusion followed full-run inspection and does not create an untouched test set. The centred convention had higher pilot agreement, but the script selected it using full-run agreement; both conventions are therefore reported. Near-surface agreement uses the selected query points within 0.01 normalized units of the raw surface, rather than all queries.

The endpoint was the within-object change in signed occupancy-volume error from offset 0 to 1.5. Percentile intervals use 5,000 paired object bootstrap resamples (seed 0). Transition-midpoint comparisons retain only objects whose recorded fit-acceptance flags pass at both endpoint offsets. These flags do not preserve failure reasons, and released-label fit acceptance was not saved; no new accepted-fit comparison with released-label midpoints is asserted.

H. Real EM processing

MitoEM-R analysis uses labels on slices 400–499 at $30 \times 8 \times 8$ nm spacing in z , y , and x . Eligible objects contain

at least 2,000 voxels, span at least three z planes, do not touch slab or image boundaries, and have a padded region of interest of at most 2,000,000 voxels. Of 1,203 instance IDs present in the slab, sequential filtering removed 586 for slab-boundary contact, 61 additional IDs for image-boundary contact, none further for size or z span, and six further for ROI size, leaving 550 (45.7%). Counts depend on this stated order; overlapping criterion failures are supplied in the reproducibility package. All 550 were meshed by marching cubes [31] at level 0.5. Mesh volume and area are compared with voxel-count volume and exposed-voxel-face area. Surface point sampling uses 10,000 points and five repetitions per object. All representations derive from the same annotation, so this is a processing-portability experiment.

In E7, masks are downsampled in y and x only, by block occupancy of at least one half with ties assigned to foreground. Spacings become $30 \times 16 \times 16$ and $30 \times 24 \times 24$ nm; z is unchanged. The 24 nm in-plane spacing matches the reported 3DMSL mesh-generation spacing; the complete voxel geometry remains $30 \times 24 \times 24$ nm [16]. Ties can occur only for the 2×2 blocks of the 16 nm grid; the 3×3 blocks of the 24 nm grid cannot tie. Remeshed volume, surface area, and sphericity, $\pi^{1/3}(6V)^{2/3}/A$, are compared with the native mesh. The experiment measures sensitivity to this regridding and meshing procedure, not the full effect of lower-resolution acquisition. Post-audit sensitivity checks enumerate all four in-plane block phases at factor two, with ties assigned either to foreground or background, and all nine phases at factor three. The original phase begins at the lower bound of each cropped object ROI. Foreground component counts are compared using both 6- and 26-neighbor connectivity; Euler characteristics use 26-neighbor foreground connectivity.

I. Statistical reporting

We report signed bias, MdAPE, RMSE, Bland–Altman limits of agreement, and pass rates at candidate margins. We use percentile bootstrap intervals [32]. Frozen evaluation uses 1,000 object-level resamples (seed 790); internal validation uses 2,000 resamples (seed 2026 with method-specific offsets). These intervals describe object-level variability and do not capture between-animal uncertainty. The primary reported uncertainty bound is the saved 2.661% development-calibrated bound of the occupancy-only model, retained without recalibration on the second batch. The 2.17% margin is the median pairwise repeatability coefficient from E2 and is retained as a descriptive sensitivity threshold, not a single-estimate accuracy bound; 5%, 6%, and 10% are sensitivity thresholds. None is a biological equivalence margin. Subgroup coverage intervals use the 95% Wilson method as an object-independence approximation. Grid-phase, topology, and mechanistic-scale checks were added after the primary results and are exploratory.

V. RESULTS

A. High overall agreement coexists with systematic representation bias

All 2,720 development objects were retained. The audit flagged 73 objects for low occupancy fraction, four for a raw

point-cloud extent discrepancy above 10% on at least one axis, and two for incomplete rendered images. Volume ICC was 0.9939, while mean signed occupancy error was +3.665% (bootstrap interval +3.572% to +3.755%) and median signed error was +3.052%. Occupancy volume exceeded mesh volume for 2,719 of 2,720 objects.

The ICC largely reflects the 68-fold range of mesh volumes. Within volume deciles, the median ICC was 0.8570 (range 0.747–0.983). Bland–Altman analysis gave a bias of +1,369 cubic mesh units (+4.67% of the mean mesh volume), 95% limits of agreement of $-3,568$ to $+6,306$, a percentage error of 16.47% using the mean of both methods (16.85% using mean mesh volume), and a proportional-bias correlation of 0.82; the absolute discrepancy grows with object size (Fig. 1). Pooled absolute limits summarize this sample and should not be interpreted as size-independent prediction limits. Occupancy extents were nearly unbiased (-0.31% to -0.03% by axis), whereas trimmed point-cloud extents were smaller than mesh extents by 1.50% to 2.31%, the expected effect of discarding extremal points.

In E2, the median estimated full-sample sampling SD was 0.784% (interquartile range 0.508%–1.088%), giving a median repeatability coefficient of 2.172%. The median signed error in this 400-object subset was +2.930% at 100,000 queries, about 3.7 times the median sampling SD. These results support a systematic component beyond estimated numerical sampling variability. The illustrative single-estimate 95% half-width at the median SD is 1.54%. Under the binomial query model, sampling SD is occupancy-dependent: at $p = 0.02$ and 100,000 queries it is 2.21%, with larger relative SD at lower occupancy.

B. An outward label-boundary offset is observed

Of the 60 fits, 57 met every acceptance criterion. The other three stopped with an optimizer termination warning but were not on a bound, and their midpoints (0.00263–0.00309) lie within the range of the accepted fits; they are excluded from primary summary statistics. For the accepted fits, the median label-transition midpoint was $t_{50} = 0.00304$ normalized units outside the mesh (interquartile range 0.00287–0.00315; coefficient of variation 0.09), with a median transition width of 0.00027. The median volume-implied offset was 0.00312, a ratio of 0.98, and the per-object Pearson correlation between the two was 0.58 (Spearman 0.35). As an exploratory influence check, omitting object 25660, which has the largest volume-implied displacement, reduces Pearson r from 0.577 to 0.312 and Spearman ρ from 0.351 to 0.315. This is not an exclusion criterion; the object remains in all primary summaries. The 32-centroid approximation returned the exact distance for every selected point. No point more than ten transition widths inside the boundary was labeled outside, and at most 0.1% of points more than ten widths outside were labeled inside, supporting a displaced label transition with few far-from-transition inconsistencies (Fig. 2).

The measured offset is 0.93 times the half-shift heuristic (0.00326) and 0.47 times the full shift (0.00651). This is consistent with inflation during label generation, including

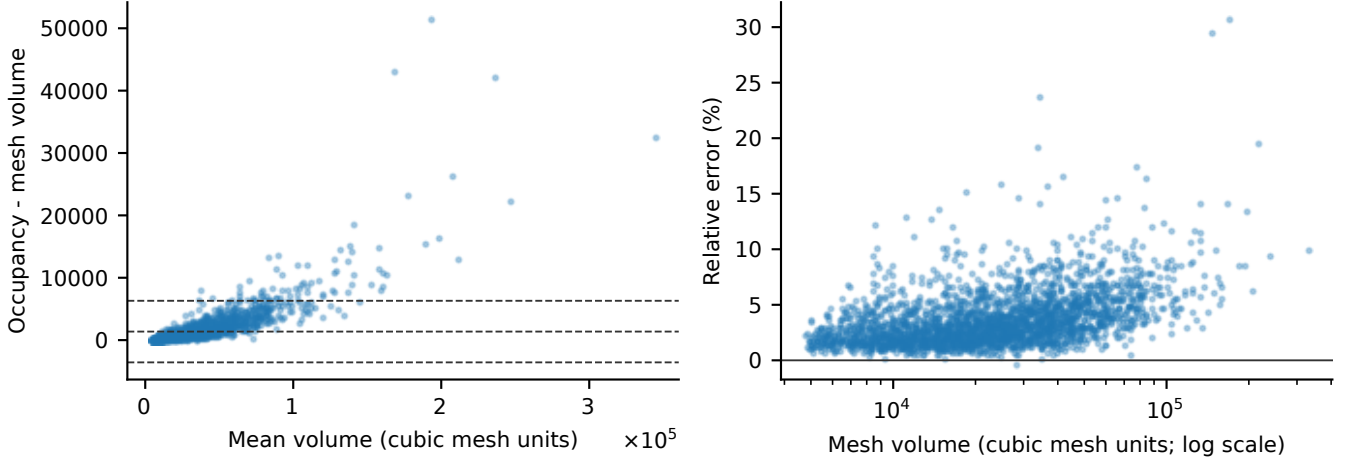


Fig. 1. Development-batch agreement diagnostics for all 2,720 objects. Left: absolute differences versus pair means, with pooled bias and limits of agreement. Right: relative errors versus mesh volume. The size-dependent pattern limits interpretation of pooled absolute limits as universal error bounds.

depth-offset watertighting, but does not isolate the responsible operation. The ratio should not be read as quantitative agreement: in the controlled reimplementations (Section V-D), the offset intervention shifted the fitted midpoint by a mean of 0.001264 normalized units among 53 accepted non-pilot pairs. Residual displacement at zero offset was not decomposed among the remaining implementation steps. A local replay reproduced the 57 originally accepted fits’ summary values; all 60 optimizations reported convergence in that environment. We retain the archived 57-object acceptance set for the primary summary; originally rejected fits are additionally marked in the figure. Including all 60 locally converged fits gives median $t_{50} = 0.00304$ and Pearson $r = 0.57$, leaving the central interpretation unchanged.

C. The frozen correction transfers with reduced coverage

Table IV compares internal and frozen second-batch performance. The occupancy-only correction reduced internal MdAPE from 3.035% to 0.580%. On the 2,728 new objects, it reduced MdAPE from 3.481% to 0.664% and RMSE from 5.743% to 1.743%, reductions of 80.9% and 69.7%. Its second-batch mean signed bias was +0.006% (95% bootstrap interval -0.059% to $+0.076\%$), and the MdAPE interval was 0.636%–0.700%.

A near-zero mean residual does not imply uniformly accurate estimates. Coverage of the occupancy-only bound fell to 92.1% (bootstrap interval 91.1%–93.0%). Corrected second-batch pass rates at 2.17%, 5%, 6%, and 10% were 88.5%, 98.0%, 98.8%, and 99.7%. Fig. 3 shows the complete absolute-error distributions, including the tails; the vertical line marks the occupancy-only bound. The uncorrected and global methods have identical coverage in both evaluations because they cover the same objects. The global correction divides every estimate by $k = 1 + c/100$, where $c = 3.65\%$ is the mean training error. It therefore covers raw errors from $c - b_g k = -1.15\%$ to $c + b_g k = 8.451\%$. The upper limit equals the uncorrected bound, because both calibration quantiles fall

TABLE IV
CORRECTION PERFORMANCE. ERRORS ARE PERCENTAGES RELATIVE TO MESH VOLUME. COVERAGE USES EACH METHOD’S OWN CALIBRATED ABSOLUTE-ERROR BOUND.

Method	Bias	MdAPE	RMSE	Bound	Cov.
<i>Internal test, $n = 550$</i>					
Uncorrected	+3.608	3.035	4.286	8.451	96.4%
Global correction	−0.040	1.370	2.231	4.632	96.4%
Occupancy-only	−0.032	0.580	1.149	2.661	96.2%
<i>Frozen second batch, $n = 2,728$</i>					
Uncorrected	+4.497	3.481	5.743	8.451	90.0%
Global correction	+0.817	1.503	3.542	4.632	90.0%
Occupancy-only	+0.006	0.664	1.743	2.661	92.1%

on the same calibration object, and no raw error fell below the lower limit.

D. Depth-offset intervention in the recreated pipeline

After pilot exclusion, every one of the 55 objects increased in occupancy-volume error at both offset steps (Table V). Under the centred convention, the mean paired increase from offset 0 to 1.5 was 1.571 percentage points (95% bootstrap interval 1.374–1.773). The literal convention gave 1.593 points (1.393–1.806), also positive in all 55 objects. The paired fused-mesh volume-error increase was 1.477 points (1.297–1.658). These interventions support a depth-offset contribution within the recreated pipeline; the remaining positive error at offset zero also shows that the offset does not account for all discrepancy.

Failure sets differed between offsets: 53 centred and 52 literal endpoint pairs passed both fit checks. Among accepted pairs, the mean midpoint increase was 0.001264 normalized units (95% interval 0.001219–0.001311) for centred and 0.001289 (0.001243–0.001336) for literal; all accepted changes were positive. Volume endpoints retain all 55 objects independently of transition-fit acceptance. All-60 sensitivity estimates were similar: mean paired volume-error increases

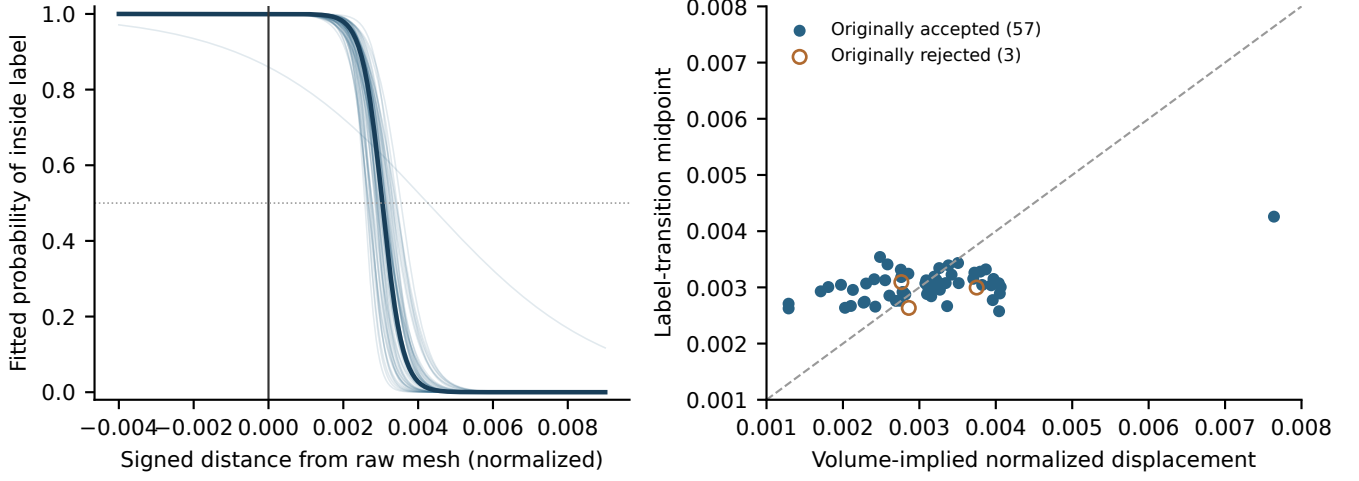


Fig. 2. Exact-distance boundary fits for the 57 objects accepted in the recorded analysis, replayed on the original queries. Left: object-specific fitted transitions (thin lines) and the transition at median fitted parameters (bold); zero marks the raw mesh. Right: label-transition midpoint versus volume-implied displacement, with identity line ($r = 0.58$). Open markers additionally show the three originally rejected fits using their local replay estimates; they are excluded from the correlation. Their convergence sensitivity is described in the text.

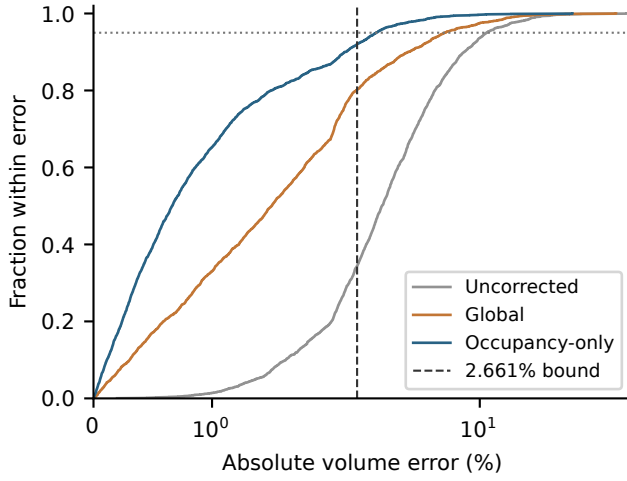


Fig. 3. Empirical cumulative distributions of absolute percentage error for all 2,728 second-batch objects. Every observation is included. The horizontal axis is linear near zero and logarithmic in the tail. The vertical line is the occupancy-only 2.661% bound; the horizontal dotted line marks 95% coverage.

were 1.579 points (1.394–1.769) for centred and 1.588 (1.400–1.783) for literal. At offset 1.5, median near-surface agreement for the non-pilot objects was 97.97% (centred) and 90.22% (literal). High agreement does not establish the historical implementation’s coordinate convention. In the ten-object hybrid check, the field-band-restricted containment replacement agreed with interpolated labels for a median of 99.4% (offset 0) and 99.7% (offset 1.5) of sampled near-surface queries. It changed occupancy-volume error by a median of +0.130 and +0.019 percentage points, with maximum absolute differences of 0.264 and 0.100, respectively. The mean paired 0-to-1.5 increase was 1.566 points with hybrid labels versus 1.683 with interpolation. At offset 1.5, median agreement with

TABLE V
EXPLORATORY OFFSET INTERVENTION AFTER RETROSPECTIVE PILOT EXCLUSION ($n = 55$). ERRORS ARE SIGNED PERCENTAGES RELATIVE TO RAW MESH VOLUME. FIT COUNTS CONCERN BOUNDARY MIDPOINTS, NOT VOLUME ELIGIBILITY.

Depth offset (voxels)	0	0.75	1.5
Median error, centred (%)	1.578	2.649	3.435
Median error, literal (%)	1.611	2.529	3.417
Accepted fits, centred	54/55	54/55	54/55
Accepted fits, literal	54/55	55/55	53/55

released labels on the sampled near-surface queries was 98.2% with hybrid labels and 98.1% with interpolation. The tested replacement preserved the positive mean offset effect in this subset and did not eliminate the discrepancy with released labels; it does not establish full containment equivalence.

E. Shape distribution and low occupancy define the transfer limits

The second batch contained larger and less spherical objects: median sphericity was 0.793 versus 0.8358, median $A_m/V_m^{2/3}$ was 6.098 versus 5.786, and median occupancy fraction was 0.1138 versus 0.1437. Low-occupancy prevalence increased from 73/2,720 (2.7%) to 181/2,728 (6.6%).

Mean raw error increased by 0.833 percentage points. The development regression on shape complexity predicted an increase of 0.747 points, about 90% of the observed change. The offset itself replicated: median volume-implied offsets were 0.00293 (development) and 0.00297 (second batch), with coefficients of variation of 0.28 and 0.27, and the slope of error on $A_m/V_m^{2/3}$ was 1.78 and 2.00 percentage points per unit ($R^2 = 0.67$ and 0.69). This descriptive decomposition associates most of the mean-error shift with changing shape complexity; it does not establish a causal partition. The added A_ms/V_m diagnostic gave development slope 0.3101

percentage points per unit predictor, corresponding to $t_{\text{norm}} = 0.00310$. Applied unchanged to the second batch, it yielded mean residual -0.062 percentage points and $R^2 = 0.828$. This supports the scale-aware geometric relationship without identifying the precise generation operation.

Table VI identifies much higher failure risk below occupancy 0.02, but the flag does not capture all violations: 124 of 216 second-batch failures (57.4%) occurred outside it. Unflagged coverage decreased from 97.8% internally to 95.1% in the second batch. Reweighting the internal subgroup coverages to second-batch subgroup proportions gives 94.17% coverage, versus 96.18% internally and 92.08% observed. This descriptive decomposition attributes 2.01 percentage points of the decline to subgroup composition and a further 2.09 points to within-group changes. The flag was defined during the initial audit, but its later interpretation as an applicability restriction is exploratory; observed 95.1% subgroup coverage is not a conditional guarantee. Within the second batch, coverage declined across shape-complexity deciles from 100% to 64.1%, and the highest decile remained under-corrected by a mean of $+1.45\%$. The worst 136 errors had median sphericity 0.557 and occupancy 0.017 (cohort medians: 0.793 and 0.114).

F. Real EM processing preserves volume more than surface descriptors

All 550 eligible MitoEM-R objects were meshed. The median mesh-versus-voxel volume difference was -0.19% (5th–95th percentile -0.35% to -0.10%). The median mesh-versus-voxel-face surface difference was -16.31% , illustrating the dependence of digitized-surface measurements on the area estimator [33]. The median trimmed point-cloud extent difference was -3.70% , whereas raw sampled extents differed by only -0.03% , reproducing on real EM the trimming effect seen in 3DMSL.

All meshes in the original regridding analysis remained watertight. At the original 24 nm in-plane grid phase, volume rank correlation was 1.000 to reported precision, while median sphericity increased by 11.76% (Table VII). The added phase and tie-rule checks qualify the apparent volume preservation: at 16 nm, median volume changes across four phases ranged from $+0.58\%$ to $+6.94\%$ with foreground ties, and from -5.47% to $+0.89\%$ with background ties. Across nine 24 nm phases, medians ranged from -0.22% to $+0.01\%$. Thus, the 16 nm result is particularly dependent on grid alignment and voting. At the original phase, 26-neighbor component counts changed in one of 550 objects at each spacing, and Euler characteristics changed in nine at each spacing. Watertightness does not establish topology preservation. These shifts characterize this annotation-processing procedure; they are not correction factors for cross-tissue or cross-instrument comparisons.

VI. DISCUSSION

A. Measurement agreement is a prerequisite for biological interpretation

The experiments separate three properties that are easily conflated. First, an estimator can be numerically precise while

targeting a boundary that differs from the reference mesh: sampling SD was about 0.8%, while the systematic offset was nearly four times larger. Second, a correction can remove average bias while leaving substantial errors in a morphological subgroup. Third, object rankings can remain stable while a descriptor’s absolute scale changes. The high ICC, the near-zero corrected mean bias, and the high regridding rank correlations are each informative, but none alone establishes interchangeability.

For structure-function studies, a processing-induced change in sphericity could enter a classifier as though it were a cellular response, and rank preservation could hide a shifted absolute threshold. The bias found here also has a practical implication for users of 3DMSL: occupancy labels target a boundary displaced from the released mesh. Models trained on those labels may inherit that discrepancy, although realized model error also depends on fitting, regularization, and reconstruction. For a constant normalized displacement, the first-order relative effect scales with $A_m s / V_m$.

B. Correction transfer and uncertainty transfer are separate results

The occupancy-only model performed well on typical second-batch objects and improved substantially on both raw volumes and a global correction. Observed coverage was near 95% in the unflagged group and about 49% in the flagged group. Coverage also declined within the unflagged group between evaluations, and aggregate subgroup coverage does not ensure coverage for every shape class. This pattern illustrates the distinction between marginal and conditional coverage [27]. Conditional residual scales, Mondrian calibration by occupancy group, or covariate-shift weighting [28] are natural next steps, but they require new development, and that batch could not serve as an untouched test for methods selected using its outcomes. The original correction and its reduced coverage are retained here as the primary transfer evidence.

C. Implications for Homeostatic Invariance

One potential application is Homeostatic Invariance, which we propose as a framework in which structural variability may coexist with maintained function within a bounded biological regime. The findings address a measurement prerequisite for testing that proposition; they do not test the proposition itself. A future biological stability region should distinguish structural variation that exceeds processing uncertainty from variation in independently measured function. That study would need aligned functional observations, longitudinal or perturbational designs, and replication across biological units. The present datasets contain no linked respiration, ATP, or membrane-potential measurements, so they cannot establish preserved function or a transition boundary.

VII. LIMITATIONS

The two 3DMSL batches share dataset construction and upstream provenance; disjoint IDs do not establish independent acquisitions or animals, and source-object lineage and

TABLE VI
OCCUPANCY-ONLY COVERAGE AT THE SAVED 2.661% BOUND. WILSON INTERVALS ASSUME INDEPENDENT OBJECTS AND DO NOT QUANTIFY BETWEEN-ANIMAL UNCERTAINTY.

Evaluation	Group	Covered / total	Failures	MdAPE	Coverage [95% interval]
Internal test	All	529 / 550	21	0.580%	96.2% [94.2, 97.5]
Internal test	Occupancy < 0.02	7 / 16	9	3.011%	43.8% [23.1, 66.8]
Internal test	Occupancy $\geq$ 0.02	522 / 534	12	0.551%	97.8% [96.1, 98.7]
Second batch	All	2,512 / 2,728	216	0.664%	92.1% [91.0, 93.0]
Second batch	Occupancy < 0.02	89 / 181	92	2.766%	49.2% [42.0, 56.4]
Second batch	Occupancy $\geq$ 0.02	2,423 / 2,547	124	0.625%	95.1% [94.2, 95.9]

TABLE VII
MITOEM-R IN-PLANE REGRIDDING RELATIVE TO NATIVE-GRID MESHES FOR THE SAME 550 OBJECTS. BRACKETS SHOW 5TH–95TH PERCENTILES.

Voxel spacing ($z \times y \times x$), nm	Volume change	Surface-area change	Sphericity change	Sphericity rank correlation
$30 \times 16 \times 16$	+0.98% [+0.45, +1.46]	−5.84% [−7.68, −4.72]	+6.90% [+5.68, +8.69]	0.997
$30 \times 24 \times 24$	−0.08% [−0.93, +0.41]	−10.60% [−12.14, −8.67]	+11.76% [+9.24, +13.75]	0.994

geometric equivalence were not resolved, despite no cross-batch identical mesh-file hashes. Object-level bootstrap intervals may be optimistic under spatial or biological dependence. The MitoEM-R analysis covers one rat-cortex volume and excludes boundary-truncated, small, shallow, and very large objects.

The internal split followed exploratory analysis. The boundary analysis covers 60 objects from one batch; three fits failed the original optimizer-success screen, although they converged in the local replay. Primary summaries retain the original accepted IDs. The code version used to build the released 3DMSL files has not been confirmed with the dataset authors, and the subsequent offset intervention uses a replacement implementation rather than that historical code. Its pilot exclusion was retrospective, convention selection used full-run agreement, and labels were obtained from field interpolation rather than mesh containment, although a ten-object hybrid check changed volume errors by at most 0.264 percentage points. That check retained interpolated labels outside a fused-field threshold band; full containment equivalence remains unverified. Boundary-fit failures were excluded from paired midpoint summaries, while their valid volume endpoints were retained. The model comparisons were also added after second-batch outcomes had been inspected; they do not replace the original frozen evaluation. The regridding experiment leaves z unchanged and does not model the point-spread function, segmentation uncertainty, or acquisition process of another instrument. Candidate tolerances were not prospectively established as biological equivalence margins.

VIII. CONCLUSION

Across 5,448 disjoint 3DMSL IDs, high representation agreement concealed a consistent positive occupancy-volume discrepancy, with boundary measurements identifying an outward label transition consistent with inflation during dataset generation. A controlled reimplementing additionally supported a depth-offset contribution to volume inflation in 55

non-pilot objects, with retrospective analysis decisions disclosed. An occupancy-only correction transferred to the second batch with an MdAPE of 0.664%; its calibrated bound covered 92.1% of all objects and 95.1% of objects outside a pre-existing low-occupancy flag. In 550 MitoEM-R objects, the original in-plane regridding preserved ranks while materially changing surface area and sphericity; additional grid-phase and tie-rule checks revealed procedure-dependent volume shifts and occasional topology changes. Numerical precision, representation bias, uncertainty coverage, and descriptor scale should therefore be reported separately. These results provide a computational measurement foundation for studies that pair mitochondrial morphology with function.

DATA AND CODE AVAILABILITY

3DMSL [25] and MitoEM [19] are publicly available. The accompanying reproducibility package supplies mesh/occupancy per-object tables, reconstructed development coefficients and scaling, split IDs, calibration bounds, boundary replay results, phase/tie/topology diagnostics, input and output hashes, analysis scripts, revised notebooks, and the archived and corrected Section 24 validation outputs. The added package records all 360 intervention condition rows, the 330 non-pilot rows, failed boundary fits, input hashes, and corrected model-sensitivity results. Deterministic reconstruction with the recorded seed reproduces the reported correction metrics and coverage counts. The package now includes the original frozen run, its model, split IDs, stored predictions, and protocol. The supplied model’s full SHA-256 matches the recorded hash (prefix 93b6daed), and all five protocol-listed input hashes match. Applying that model to the archived descriptors, without refitting, reproduced all 2,728 stored occupancy-only residuals to within 7×10^{-14} percentage points. Original artifacts and locally reconstructed models are stored separately. This verifies artifact consistency, not the chronology of every analytical decision. A result-to-file map identifies supporting code and outputs. The package additionally preserves the executed Section 25 notebook, its per-object

containment records, protocol, and model-comparison outputs. Independent reaggregation of all 40 object/offset/convention records, including 20 centred-convention hybrid records, exactly reproduced the saved ten-object summaries. This verifies the output aggregation, not a fresh rendering or full-containment rerun. The package distinguishes original reported results, local reproductions, and post-audit exploratory diagnostics. Reproduction covers the primary correction and selected boundary and resolution diagnostics; it is not an end-to-end rerun of every notebook analysis. No public repository DOI is claimed.

ETHICS STATEMENT

This study analyzed only publicly available, previously published datasets. No new animal experiments or human-participant research were conducted.

APPENDIX A

EXPLORATORY CORRECTION-MODEL SENSITIVITY

In E10, after the original outcomes had been inspected, we compared the retained five-feature ordinary least-squares (OLS) correction with occupancy-fraction-only OLS and ridge regression on the same five features (Table VIII). Each model used the saved development training split; residual bounds used only the calibration split. Ridge used 36 log-spaced penalties from 10^{-3} to 10^4 and five unshuffled folds within training, choosing the smallest mean validation squared error. Standardization was refit inside each fold using a scikit-learn pipeline before full-training refitting. The exploratory follow-up also changed the selection criterion from mean fold R^2 to mean squared error. With fold-specific scaling, the former selected $\alpha = 0.3981$ and the latter $\alpha = 0.2512$; the difference should not be attributed to scaling alone. The table reports the MSE-selected model. An independent local NumPy rerun on reconstructed geometry tables reproduced its displayed metrics.

Refitted retained-model residuals agreed with the stored second-batch residuals across all 2,728 rows to within 4.46×10^{-14} percentage points; the global correction agreed to 2.49×10^{-14} . This verifies numerical reproduction of table-based predictions, not a fresh end-to-end raw-data rerun. Compared with retained OLS, second-batch MdAPE was higher by 0.839 percentage points for global correction (95% paired object-bootstrap interval 0.782–0.889), 0.263 for fraction-only OLS (0.221–0.305), and 0.001 for ridge (−0.003–0.006). Intervals use 5,000 resamples with seed 0. Similar ridge and OLS MdAPE does not constitute a formal equivalence test. These exploratory comparisons support the retained specification without changing its primary frozen results or resolving reduced uncertainty coverage.

APPENDIX B

EXPLORATORY SUPPLEMENTARY IMAGING ANALYSIS (S1)

This exploratory analysis is separate from the primary structural results, and its simulator-configuration provenance remains unresolved. We tested whether 2D threshold morphology of the released 3DMSL simulated microscopy images

TABLE VIII

EXPLORATORY SECOND-BATCH MODEL COMPARISON ($n = 2,728$). MDAPE AND RMSE ARE PERCENTAGES; COVERAGE IS THE PERCENTAGE WITHIN EACH MODEL’S OWN DEVELOPMENT-CALIBRATED BOUND. RIDGE USES CORRECTED FOLD-SPECIFIC SCALING.

Correction	MdAPE	RMSE	Coverage (%)
None	3.481	5.743	90.03
Global mean	1.503	3.542	90.03
Occupancy fraction only	0.927	2.718	89.37
Retained five-feature OLS	0.664	1.743	92.08
Ridge, same features	0.665	1.747	91.97

recovers object geometry. For 300 development objects sampled without replacement with seed 790, three configurations (Conf1 confocal, 70 nm pixels; Conf2 confocal, 48.8 nm; Epi1 widefield, 109 nm) and six orientations each, we segmented half-peak signal above a border-background estimate, applied morphological closing and hole filling, and retained the largest component. The analysis processed 1,798 Conf1 views and 1,800 views each for Conf2 and Epi1, with no empty or edge-touching masks. Image areas were compared with view-specific mesh silhouettes and sections that follow the simulator’s centering and rotation; selection of the best-matching target was exploratory.

The results were negative. Geometry explained only 13%–28% of the orientation-to-orientation variation in image area. Cross-configuration ICC(A,1) for area was 0.509 for Conf1 versus Conf2, and 0.051 and 0.058 for the two Epi1 pairs; the corresponding rank correlations were 0.392, 0.344, and 0.631. Image-to-geometry area ratios were about 0.77 for the confocal configurations and 2.82 for Epi1. Rescaling the 109 nm Epi1 pixels to match the confocal ratio gives $109\sqrt{0.77/2.82} \approx 57.0$ nm; matching unit image-to-geometry agreement instead gives $109/\sqrt{2.82} \approx 64.9$ nm. These are diagnostic rescalings, not independently measured pixel calibrations. The supplied code notes a different pixel size in another repository widefield configuration, so simulator-configuration provenance remains unresolved. Blur, thresholding, focal geometry, and calibration can all contribute. All three configurations were noisy (median signal-to-noise ratio 6–18), and objects spanned only a few pixels near the optical resolution limit (124–245 nm). Bright hotspots caused the half-peak threshold to miss dimmer parts of objects, and an Otsu threshold captured background noise. For these simulated configurations and the tested half-peak and Otsu thresholds, image-derived areas did not reliably recover the candidate mesh targets. This finding does not establish a limitation of all segmentation methods or all optical acquisition conditions. These images were not used for the main analyses.

APPENDIX C

REPRODUCTION AND ADDITIONAL SENSITIVITY CHECKS

The local reproduction uses the supplied notebooks and original archives. A second-batch analysis applies only development-derived coefficients, with no fitting or recalibration on second-batch outcomes. The original frozen model and stored second-batch residuals are included. All five archived input files match the hashes in the run protocol. Direct model

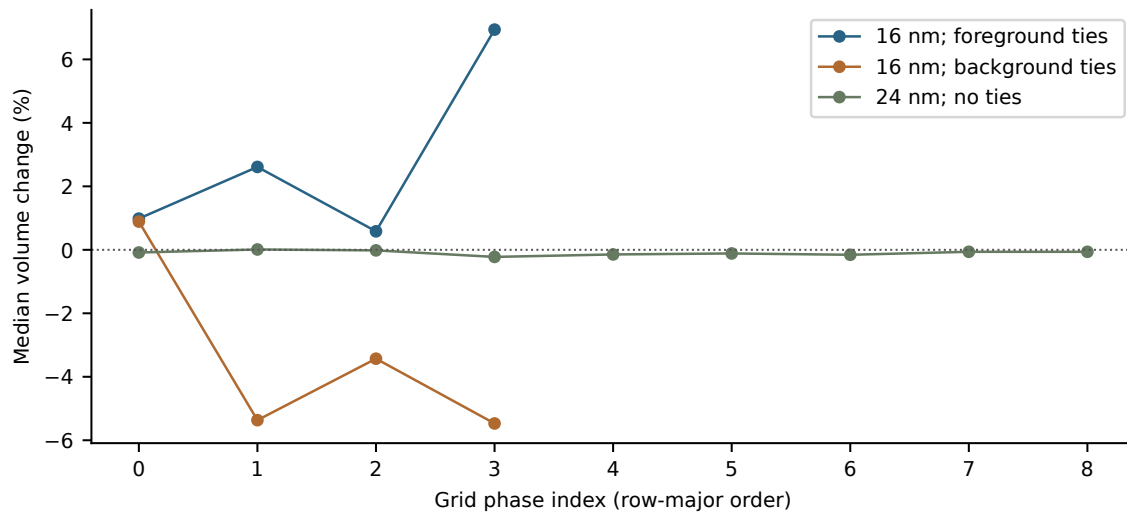


Fig. 4. Exploratory grid-phase sensitivity for all 550 eligible MitoEM-R objects. Each point is the median volume change from its native-grid mesh. Phase indices enumerate (y, x) offsets in row-major order, starting at $(0, 0)$: four phases for factor two and nine for factor three. Leading zero padding changes the block origin without cropping the object. Lines connect the discrete phase configurations for readability.

application reproduced the stored occupancy-only residuals to within 7×10^{-14} percentage points across 2,728 objects. Separately refitted models reproduce the numerical results but are retained as reconstructions, not byte-identical substitutes for the original artifact. The corrected coverage checker reads the named global-offset field and validates it against the algebra of stored residuals, rather than searching model-summary keys heuristically.

The originally implemented query-box estimator uses empirical coordinate spans. Replacing these with the nominal width 1.1s raises occupancy volume by a median 0.1066% in each batch. This sensitivity does not alter the retained estimator or saved correction. The supplement provides sampling-SD summaries by occupancy under the independent Bernoulli-query approximation.

Boundary replay retains all 60 object IDs and records original acceptance decisions separately from local optimizer status. Figure 2 distinguishes the 57 originally accepted objects from the three originally rejected fits. Phase and tie-rule checks cover all 550 eligible MitoEM-R objects in 17 configurations (9,350 remeshed object/configuration pairs). All pairs produced nonempty meshes. Their full per-object descriptor, component-count, and Euler-characteristic results are included in the package. These post-audit checks were not prospectively registered and do not replace the original correction-transfer evaluation.

The phase sensitivity in Fig. 4 shows why the original-grid volume result is procedure-specific. The boundary influence check likewise quantifies sensitivity without deleting a valid observation or redefining the primary acceptance set. All sensitivity outputs are supplied separately from the reconstructed development model.

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